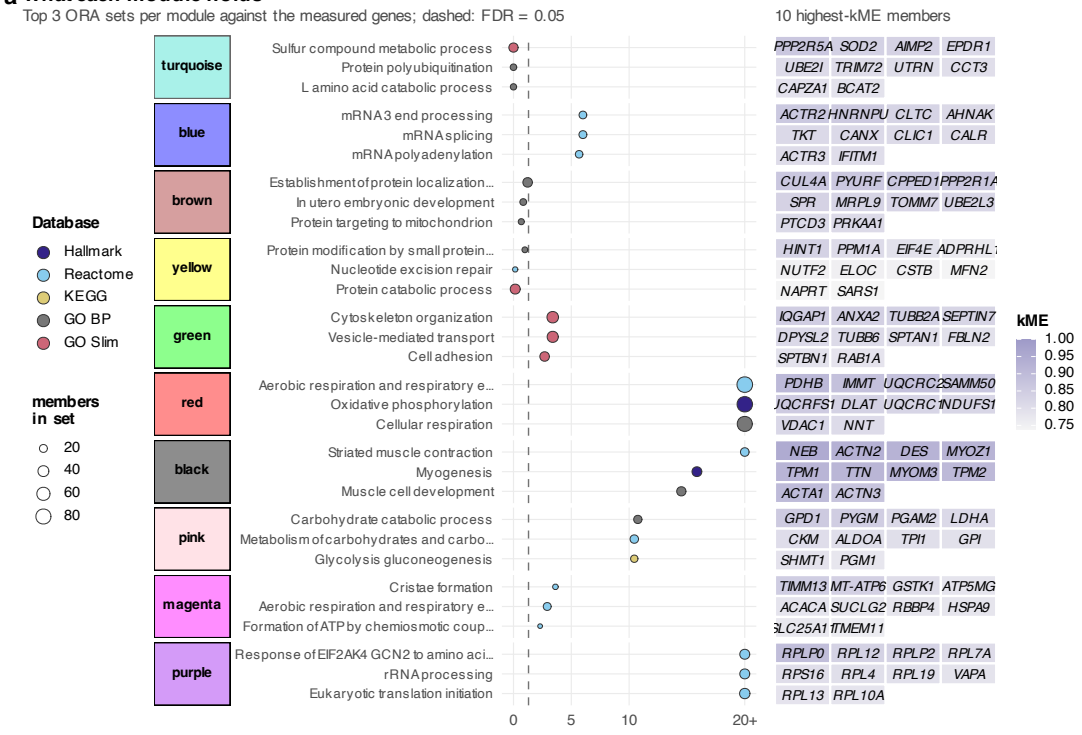
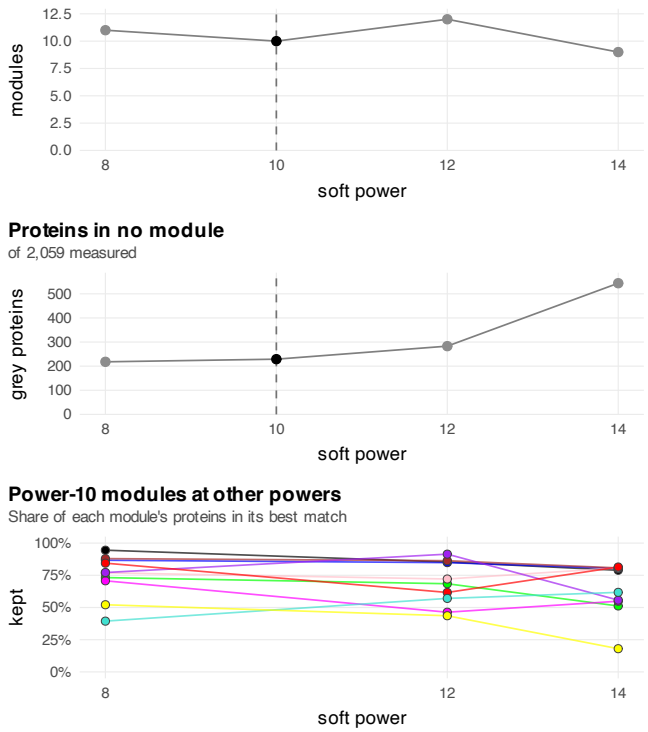
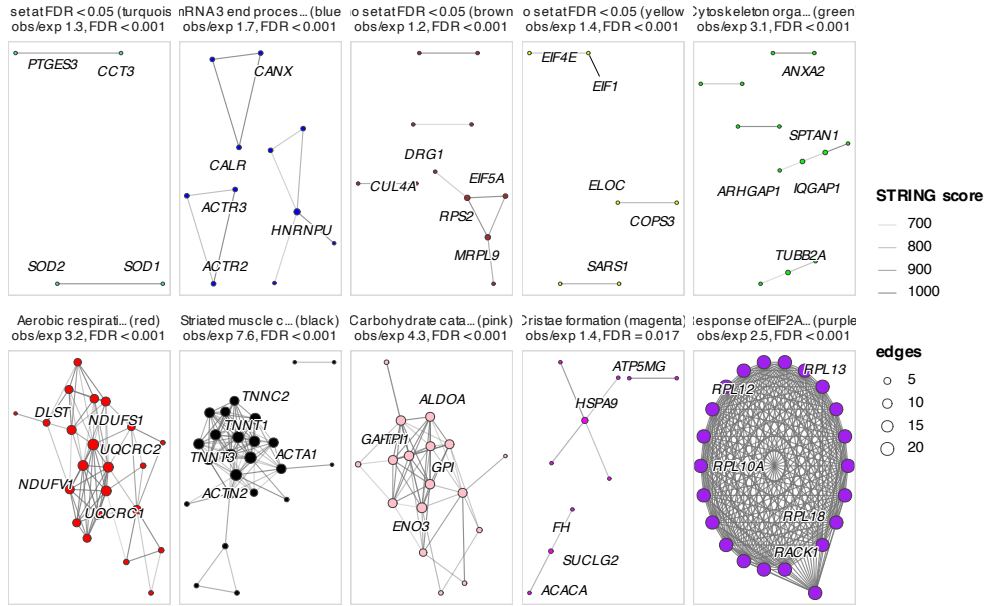
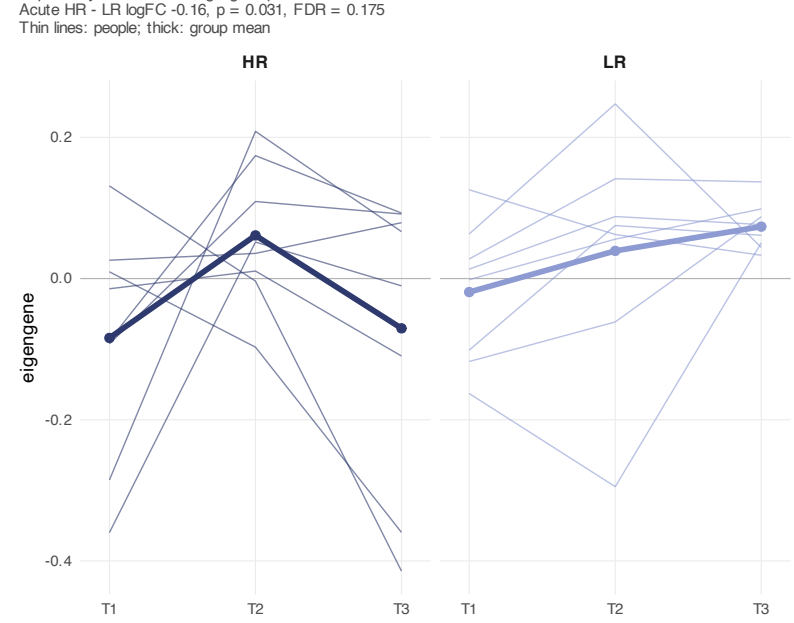

a What each module holds
Top 3 ORA sets per module against the measured genes; dashed: FDR = 0.05
10 highest-kME members
turquoise
blue
brown
yellow
green
red
black
pink
magenta
purple
Sulfur compound metabolic process
Protein polyubiquitination
L amino acid catabolic process
mRNA 3 end processing
mRNA splicing
mRNA polyadenylation
Establishment of protein localization...
In utero embryonic development
Protein targeting to mitochondrion
Protein modification by small protein...
Nucleotide excision repair
Protein catabolic process
Cytoskeleton organization
Vesicle-mediated transport
Cell adhesion
Aerobic respiration and respiratory e...
Oxidative phosphorylation
Cellular respiration
Striated muscle contraction
Myogenesis
Muscle cell development
Carbohydrate catabolic process
Metabolism of carbohydrates and carbo...
Glycolysis gluconeogenesis
Cristae formation
Aerobic respiration and respiratory e...
Formation of ATP by chemiosmotic coup...
Response of EIF2AK4 GCN2 to amino aci...
rRNA processing
Eukaryotic translation initiation
PPP2R5A SOD2 AIMP2 EPDR1
UBE2I TRIM72 UTRN CCT3
CAPZA1 BCAT2
ACTR2 HNRNPU CLTC AHNAK
TKT CANX CLIC1 CALR
ACTR3 IFITM1
CUL4A PYURF CPPED1 PPP2R1A
SPR MRPL9 TOMM7 UBE2L3
PTCD3 PRKAA1
HINT1 PPM1A EIF4E ADPRHL1
NUTF2 ELOC CSTB MFN2
NAPRT SARS1
IQGAP1 ANXA2 TUBB2A SEPTIN7
DPYSL2 TUBB6 SPTAN1 FBLN2
SPTBN1 RAB1A
PDHB IMMT UQCRC2 SAMM50
UQCRFS1 DLAT UQCRC1 NDUFS1
VDAC1 NNT
NEB ACTN2 DES MYOZ1
TPM1 TTN MYOM3 TPM2
ACTA1 ACTN3
GPD1 PYGM PGAM2 LDHA
CKM ALDOA TPI1 GPI
SHMT1 PGM1
TIMM13 MT-ATP6 GSTK1 ATP5MG
ACACA SUCLG2 RBBP4 HSPA9
SLC25A1 TMEM11
RPLP0 RPL12 RPLP2 RPL7A
RPS16 RPL4 RPL19 VAPA
RPL13 RPL10A
0
5
10
20+
-log10 FDR
Database
Hallmark
Reactome
KEGG
GO BP
GO Slim
members in set
20
40
60
80
kME
1.00
0.95
0.90
0.85
0.80
0.75


c Modules per soft power
Power 10 used (dashed)
modules
soft power
Proteins in no module
of 2,059 measured
grey proteins
soft power
Power-10 modules at other powers
Share of each module's proteins in its best match
kept
soft power


b STRING interactions within modules
Edges (combined score ≥ 700) among each module's 25 highest-kME members; the 5 best-connected named. Strip: module-wide STRING enrich
set at FDR < 0.05 (turquois obs/exp 1.3, FDR < 0.001
nRNA 3 end proces... (blue obs/exp 1.7, FDR < 0.001
o set at FDR < 0.05 (brown obs/exp 1.2, FDR < 0.001
o set at FDR < 0.05 (yellow obs/exp 1.4, FDR < 0.001
Cytoskeleton orga... (green obs/exp 3.1, FDR < 0.001
PTGES3
CCT3
SOD2
SOD1
CANX
CALR
ACTR3
HNRNPU
ACTR2
DRG1
CUL4A
EIF5A
RPS2
MRPL9
EIF4E
EIF1
ELOC
COPS3
SARS1
ANXA2
SPTAN1
ARHGAP1
IQGAP1
TUBB2A
Aerobic respirati... (red) obs/exp 3.2, FDR < 0.001
Striated muscle c... (black) obs/exp 7.6, FDR < 0.001
Carbohydrate cata... (pink) obs/exp 4.3, FDR < 0.001
Cristae formation (magenta) obs/exp 1.4, FDR = 0.017
Response of EIF2A... (purple obs/exp 2.5, FDR < 0.001
DLST
NDUFS1
UQCRC2
NDUFV1
UQCRC1
TNNC2
TNNT1
TNNT3
ACTA1
ACTN2
ALDOA
GAPDH
TPI1
GPI
ENO3
ATP5MG
HSPA9
FH
SUCLG2
ACACA
RPL13
RPL12
RPL10A
RPL18
RACK1
STRING score
700
800
900
1000
edges
5
10
15
20


d Cristae formation (magenta), person by person
Exploratory: the smallest eigengene p outside the interactions
Acute HR - LR logFC -0.16, p = 0.031, FDR = 0.175
Thin lines: people; thick: group mean
HR
LR
eigengene
T1
T2
T3